Triadic Casimir — PKLSA v0.3.5 Framework Report

SST Workbench / Unregistered instance

October 19, 2026

1 Instance identity

A065-v0.3.7 versioned Triadic-Casimir research instance.

2 Research question

Does a preregistered triadic clock/stress geometry pilot satisfy its analytic controls while independent physical coupling gates remain unresolved?

3 Null and alternative hypotheses

No independently qualified source-specific physical two-surface response is established by the present geometry and toy models.

4 Explicit assumptions

Time-dependent regularized filament RK4 is exploratory and is not full Euler closure.

5 Definitions

u material velocity; omega=curl u; S symmetric strain.

6 Analytical equations

The official AUTO_SCIENCE_CONTRACT fragment records frozen equations.

Machine-readable science contract

Research question Does a preregistered triadic clock/stress geometry pilot satisfy its analytic controls while independent physical coupling gates remain unresolved?

Objective Independently verify three dataset seals, analytic Euler/gauge/sheet controls and one sourced KnotPlot trefoil finite-core geometric response without importing target force.

H_0 No independently qualified source-specific physical two-surface response is established by the present geometry and toy models.

H_1 Independently qualified PKLSA Euler motion, boundary conditions and microscopic link coupling produce held-out physical predictions.

Registered equations

E1: Incompressible Euler kinematics

$$\nabla \cdot \mathbf{u} = 0, \quad \boldsymbol{\omega} = \nabla \times \mathbf{u} \tag{1}$$

Dimensional check: divergence and vorticity both inverse seconds

E2: Pressure Poisson source

$$\nabla^2 p = \rho_M(\frac{1}{2}|\omega|^2 - ||S||_F^2) \tag{2}$$

Dimensional check: kg/(mˆ3 sˆ2) on each side

E3: Regularized geometric velocity diagnostic

$$\mathbf{u}(\mathbf{x}) = \frac{1}{4\pi}\sum_j \frac{\Delta\mathbf{X}_j \times (\mathbf{x} - \mathbf{X}_{j+1/2})}{(|\mathbf{x} - \mathbf{X}_{j+1/2}|^2 + \varepsilon^2)^{3/2}} \tag{3}$$

Dimensional check: dimensionless Gamma=1 geometry diagnostic

E4: Profile-dependent cross-sectional enstrophy inverse

$$Z_G = \Gamma^2/(2\pi a^2), \quad Z_R = \Gamma^2/(\pi a^2) \tag{4}$$

Dimensional check: Gammaˆ2/Z = Lˆ2 for SI cross-section

E5: Rosenhead regularization swirl and steady radial pressure balance

$$v_{RM} = \Gamma r/(2\pi(r^2 + \varepsilon^2)), \quad \partial_r p = \rho v_\theta^2/r \tag{5}$$

Dimensional check: pressure gradient M/(Lˆ2 Tˆ2)

Registered code-to-mathematics steps

1. **S1** [G1; equations E1]: Verify all three source seals before any scientific result is evaluated
2. **S2** [G2; equations E2, E3]: Evaluate analytic Euler controls and preregistered trefoil geometry resolution scan
3. **S3** [G3; equations E3]: Withhold physical Casimir claims without independent providers, wall dynamics, and atom-link normalization
4. **S10** [G15; equations E3]: Verify direct parametric XYZ N600 to N4800 hashes and nesting
5. **S11** [G16; equations E3]: Independent fixed-core 1 percent spatial convergence at baseline epsilon=0.12 across BOTH historical trefoils
6. **S12** [G17; equations E3]: Report eps 0.06 0.12 0.24 numerical core sensitivity without physical force normalization
7. **S13** [G18; equations E3]: Check C++17 1200 and 4800 time step refinement and Python N150 parity
8. **S7** [G12; equations E3]: Audit historical Fourier and torus XYZ hash and analytic geometry
9. **S8** [G13; equations E3]: Compare reduced vortex filament RK4 spatial/time resolution on independent geometric seeds
10. **S9** [G14; equations E3]: Report fixed core regularization sensitivity without physical force interpretation
11. **S4** [G10; equations E3]: Verify fixed-core 150/300/600 convergence, separately retaining original G2 failures
12. **S6** [G3; equations E1, E2, E3]: Evaluate G3 only after precommitted G1+G10 and temporal/analytic guard; retain G2=FAIL and physical UNRESOLVED
13. **S5** [G11; equations E3]: Verify exact Fremlin raw hash and native C++ time-series refinement; never infer real Euler solution
14. **S19** [G19; equations E3]: Regenerate 2015 parametric trefoils inside falsifier, hash-check vs pinned XYZ and original generator
15. **S20** [G20; equations E3]: Measure mean segment length divided by fixed regularization epsilon at each resolution without fitting

16. **S21** [G21; equations E3]: Test native C++ RK4 fine-mesh spatial convergence for regenerated trefoils at epsilon=0.06
17. **S22** [G22; equations E4, E5]: Verify precommitted finite-core analytic source
18. **S23** [G23; equations E4, E5]: Invert independently specified Gamma and Z
19. **S24** [G24; equations E4, E5]: Hold out vortex and pressure-gradient comparisons, no cutoff refit
20. **S25** [G25; equations E4, E5]: Test mapping nonuniqueness
21. **S26** [G26; equations E4, E5]: Verify native FP64 kernel parity
22. **S27** [G27; equations E4, E5]: Keep physical core unqualified without external vorticity cross section

7 Scientific algorithm

Verify three seals; compute controls; evaluate trefoil diagnostic.

8 Dataset provenance

D01 D02 D03, KnotPlot N300, external Fremlin and separate 2015 historical Fourier and torus geometric seeds. No historical geometry is physical PKLSA evidence.

9 Independent gates

G12-G14 remain historical tests; G15-G18 separately qualify direct N600-N4800 geometries and reduced ODE, independent of physical G3. No physical conclusions.

10 Numerical controls

Dimensionless finite-segment Biot-Savart and finite-difference derivatives.

11 Backend authority

CPU profile enables strict C++ gate; G3 may be evaluated but physical validation remains unresolved.

12 Blindness and integrity

All three dataset SHA256 seals checked; reveal comparator not used in blind kernel.

13 Statistics

Geometry subsamples are not independent upstream providers.

14 Result status

Read dynamic AUTO_RESULTS fragment; no target-matched force.

Automatic run provenance

Field	Recorded value
Protocol bundle SHA-256	d24a0b62b9b439429091e256e9245870afb069cea0c534e3c90e1df16400719a
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0

Output manifest SHA-256	fbdd6b6de83205a3aad3245ccf09749de8a334f5a285c13697530d80a8cfd47a

Gate ledger

Gate	Status	Question
G0	PASS	Is the frozen protocol intact and is the runtime self-consistent?
G1	PASS	Are all preregistered sources present, identified and provenance-qualified?
G2	FAIL	Do reference numerics satisfy analytic/sanity and dimensional checks?
G10	PASS	Does refined KnotPlot spatial RK4 filament discretization pass 150/300 and 300/60
G11	FAIL	Does authentic Fremlin C++17 provider satisfy source, spatial and time checks (sin
G12	PASS	Are historical Fourier and torus XYZ inputs hash-valid and analytically consistent?
G13	FAIL	Do both historical geometries satisfy preregistered reduced filament ODE resolution
G14	PASS	Were epsilon sensitivity diagnostics recorded without fitting physical claims?
G15	PASS	Are direct-analytic historical 600/1200/2400/4800 XYZ hash and nesting controls v
G16	PASS	Do BOTH historical trefoils meet 1 percent spatial convergence at baseline epsilon (
G17	PASS	Are all three precommitted finite-core epsilon scans and complete 4800 point diagno
G18	PASS	Does native C++17 RK4 meet temporal convergence at 1200/4800 and Python spo
G19	PASS	Does the on-run analytic trefoil generator reproduce sealed XYZ and its arclength p
G20	PASS	Are N-dependent segment-to-epsilon ratios computed and declared without fitting a
G21	PASS	Do fresh formula-generated N1200/2400/4800 trefoils converge at fixed epsilon 0.06
G22	PASS	Do the synthetic finite-vorticity source invariants, preregistration and immutable leg
G23	PASS	Can profile-conditional finite core width a be independently recovered from Gamma
G24	FAIL	Does one slope-matched Rosenhead epsilon predict held-out swirl AND pressure-gra
G25	PASS	Do distinct legitimate kernel-equivalence mappings imply materially distinct epsilon
G26	PASS	Does an actually compiled C++17 FP64 radial-kernel path match Python, kernel-o
G27	UNRESOLVED	Is a measured, independent full-PKLSA transverse vorticity core and separate held-
G5	PASS	Does actual C++17 FP64 filament velocity match Python at 75/150/300? Kernel-o
G3	UNRESOLVED	Does target-free discovery reject the preregistered null under the registered statistic
G4	NOT_RUN_PREREQUISITE	Does an independent held-out/provenance-family confirmation reproduce the discov
G6	NOT_APPLICABLE	If enabled, does GPU screening pass CPU to GPU parity without being promoted b
G7	NOT_APPLICABLE	Does the result replicate across preregistered source families/encodings without pseu
G8	NOT_RUN_PREREQUISITE	Are physical PKLSA Euler, material, surface and independent-provider closure requ
G9	DEFERRED	After all blind gates, does reveal verification match the preregistered commitment a

Backend manifest


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{
  "native_kernel": {
    "authority": "KERNEL_ONLY_NOT_FRAMEWORK_G5",
    "backend_executed": "cpp17-serial-fp64",
    "comparisons": 1575,
    "compiler": "g++",
    "expected_component_count": 1575,
    "max_abs_component_error": 0.0,
    "max_relative_component_error": 0.0,
    "meaning": "Velocity-kernel parity only. NOT full RK4, Euler PDE, physical G5 or PKLSA dynamics.",
    "requested_backend": "cpp-17-binary64",
    "rms_component_error": 0.0,
    "schema": "A065-CPP-KERNEL-QUALIFICATION-1",
    "source_sha256": "11bec24153e8471f94e43dfeb0bae7b675fcb84a240e7d72092d08da7528fabd",
    "status": "PASS",
    "threshold": 5e-11
  },
  "physical_force_status": "UNRESOLVED",
  "pklsa_geometry": "a60b1a95a00110e3cb32087e7b4328d16e967379ebf97d60117b0ec05692ffb2",
  "provider_manifest_sha256": "74cc9349324a15f4c1d4ca1a6910845af45aff69512538ec362ea4ae674383dc",
  "provider_status": "PASS_RAW_HASH_AND_XYZ_PARSER_ONLY",
  "python_reference": {
    "actual_backend": "python-stdlib-binary64",
    "authority": "REFERENCE",
```

```
    "precision": "float64"
  },
  "schema": "SST-BACKEND-MANIFEST-2",
  "scope": "Kernel-only native gate; complete coupled physical validation withheld",
  "sealed_source_groups": [
    "D01_EULER",
    "D02_CLOCK_LINK",
    "D03_TWO_SURFACE"
  ],
  "v033_numerical_routing": {
    "expected_failure_ids": [
      "D01_G08_FILAMENT_SPATIAL_REFINEMENT",
      "D01_G10_FILAMENT_SPATIAL_75_300"
    ],
    "g10_status": "PASS",
    "g3_permitted_outcome": "UNRESOLVED",
    "historical_G2": "FAIL_RETAINED",
    "observed_failure_ids": [
      "D01_G08_FILAMENT_SPATIAL_REFINEMENT",
      "D01_G10_FILAMENT_SPATIAL_75_300"
    ],
    "physical_force_status": "UNRESOLVED",
    "prereg_sha256": "3856a587779d82df3b11ad1e64c4e4a45b6b4318e30c00676e4015d4df841fd6",
    "protected_mismatches": [],
    "ready_for_G3_evaluation": true,
    "reference_count": 19,
    "release": "v0.3.3",
    "schema": "A065-ROUTING-DECISION-V033-1",
    "scope": "Execution-eligibility diagnostics only. NOT Casimir, gravity, full Euler or PKLSA physical
    "time_discretization": "PASS"
  },
  "v034_historical": {
    "gate_scope": "GEOMETRY_SIMULATION_ONLY",
    "native_status": "EXECUTED",
    "prereg_sha256": "73545eaddc62a3f2eda0553610220812758dc628ad2aa0eb8d4e9177d36207e5"
  },
  "v035_highres": {
    "native_status": "EXECUTED",
    "prereg_sha256": "2e36f33e328020fd0942d89573fda8a774f6e41e1e2a8245ef46edf33f23113a",
    "scope": "DIMENSIONLESS_FILAMENT_ODE_ONLY"
  },
  "v036_generated": {
    "native_status": "EXECUTED",
    "prereg_sha256": "7ca11af33d2e48fdc3293b6153e76ce97be403b250cf54afe382fe89518a15bd",
    "scope": "PARAMETRIC_GEOMETRY_SIMULATION_ONLY"
  },
  "v037_finite_core": {
    "G24": "FAIL",
    "G27": "UNRESOLVED_NO_QUALIFIED_PHYSICAL_CORE_PROVIDER",
    "prereg_sha256": "486414f5da90661ca5e13123f44290a3651ccac26ab71d260591656947f88429",
    "scope": "SYNTHETIC_PROFILE_ONLY"
  }
}
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Source manifest


```
{
  "manifest_sha256": "f5935ef7fff71d46cda065ca0edbab713b5190a90a74d7c25fe25d8b996d5c75",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "A046",
```

```
  "evidence_class": "synthetic_control",
  "exists": true,
  "independence_family": "synthetic-euler",
  "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A065_Triadic_Casimir_B
  "provenance_family": "synthetic-euler",
  "registry_hint": {
    "catalog_id": "A046",
    "hits": []
  },
  "role": "analytic_control",
  "sha256": "9c4da362ab58a546fe8aff349a024dcc7fb63218ef30053dceee3d6659c574ad",
  "source_id": "D01",
  "version": "v0.3.7"
},
{
  "catalog_id": "E010",
  "evidence_class": "simulation",
  "exists": true,
  "independence_family": "knotplot-relaxed-3_1",
  "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A065_Triadic_Casimir_B
  "provenance_family": "knotplot-relaxed-3_1",
  "registry_hint": {
    "catalog_id": "E010",
    "hits": []
  },
  "role": "geometry_diagnostic",
  "sha256": "a60b1a95a00110e3cb32087e7b4328d16e967379ebf97d60117b0ec05692ffb2",
  "source_id": "PKLSA",
  "version": "v0.3.7"
},
{
  "catalog_id": "CANON",
  "evidence_class": "synthetic_control",
  "exists": true,
  "independence_family": "synthetic-link",
  "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A065_Triadic_Casimir_B
  "provenance_family": "synthetic-link",
  "registry_hint": {
    "catalog_id": "CANON",
    "hits": []
  },
  "role": "toy_control",
  "sha256": "8e4bcfcdf186e539f0cdeafd9cbbf9f31dbe7bbfb95c14d89099e6c31f760af8",
  "source_id": "D02",
  "version": "v0.3.7"
},
{
  "catalog_id": "CANON",
  "evidence_class": "synthetic_control",
  "exists": true,
  "independence_family": "synthetic-sheet",
  "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A065_Triadic_Casimir_B
  "provenance_family": "synthetic-sheet",
  "registry_hint": {
    "catalog_id": "CANON",
    "hits": []
  },
  "role": "heldout_control",
  "sha256": "fff8860c2597840d0f15906a43f8710793c0909ebfdde1b1eb6f64aea7863adc",
  "source_id": "D03",
  "version": "v0.3.7"
}
```

```
  ],
  "workbench_root": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A065_Triadic_Cas
}
```


15 Scientific interpretation

Physical Casimir force and gravitation remain unvalidated.

16 Reproduction

Run native FREEZE and FULL via the installation bridge.